

JISHNU SURESH

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PROFESSIONAL SUMMARY

Computer Science (Data Science) undergraduate with hands-on experience in backend engineering, REST API development, SQL database systems, and AI/LLM tooling. Built production-style tools spanning SQL migration engines, distributed data pipelines, and local LLM evaluation frameworks. Focused on AI infrastructure, developer tools, backend systems, and performance engineering.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, Bash, Verilog, Assembly

Frameworks: Flask, Flask-RESTful, SQLAlchemy, Flask-JWT-Extended

Databases: MySQL, PostgreSQL, SQLite, HBase

Big Data: Hadoop, MapReduce

AI / ML Tools: LM Studio, llama.cpp, Ollama, Scikit-learn, Pandas, NumPy

Developer Tools: Git, GitHub, VS Code, IntelliJ IDEA, DataGrip, Jupyter Notebook, Google Colab, Postman, curl, repomix

Operating Systems: Linux, macOS, Windows

PROJECTS

SQL Database Migration Tool | Python, Flask, SQLAlchemy, MySQL, PostgreSQL, SQLite, JWT

Web-based database migration platform supporting schema conversion and data migration across MySQL, PostgreSQL, and SQLite dialects.

- Engineered 10+ REST API endpoints covering authentication, migration execution, schema generation, import/export, migration history, and admin statistics.
- Designed a cross-dialect SQL schema conversion engine supporting migration of 155+ table schemas between MySQL, PostgreSQL, and SQLite.
- Implemented JWT-based authentication system using Flask-JWT-Extended with login, registration, logout, and role-based user management.
- Built SQL dump parser supporting full import/export workflows for cross-database data portability and backup/restore operations.
- Automated migration status tracking and history logging to improve debuggability and auditability of multi-step migration pipelines.

Kino – Distributed Movie Recommendation System | Python, Hadoop, MapReduce, HBase

Distributed data pipeline for large-scale movie recommendation using the MovieLens 20M dataset (~20M ratings, 27K movies, 138K users).

- Implemented MapReduce jobs to process 20 million ratings across a distributed Hadoop cluster, enabling scalable collaborative filtering at dataset scale.
- Integrated HBase as the distributed storage layer for efficient read/write access to recommendation outputs and user-item rating matrices.
- Designed end-to-end data pipeline from raw MovieLens ingestion through distributed processing to structured recommendation output.

Nocturne – Russian Vocabulary Trainer | Python, Excel, Git | github.com/Jishnu-JSK/nocturne-ru-vocabulary-trainer

Lightweight Python CLI application for structured foreign language vocabulary learning, backed by an Excel dataset of 1,000+ vocabulary entries.

- Designed an Excel-driven vocabulary engine enabling structured learning and revision sessions without requiring a database backend.
- Implemented interactive practice session logic supporting memorization workflows with clean separation of data and application logic.
- Architected for extensibility — vocabulary dataset expandable by updating a single Excel file with zero code changes required.

INTERNSHIP EXPERIENCE

Python Developer Intern

Dinesh Information Technology Systems (DITS), Kannur

June 2026 – July 2026

- Developed and integrated 10+ Flask REST API endpoints for a production SQL migration tool, handling authentication, migration execution, and admin workflows.
- Engineered SQL dump import/export functionality supporting MySQL, PostgreSQL, and SQLite, validated against schemas containing 155+ database tables.
- Debugged and resolved critical Flask API issues including frontend-backend payload mismatches, database connection failures, and migration edge cases.
- Extended database support by adding SQLite compatibility, enabling lightweight local testing and broadening the tool's supported migration targets.
- Wrote pytest-based backend tests for migration pipelines and performed systematic API validation using Postman and curl.
- Improved migration history and status tracking system to enhance pipeline observability and simplify debugging of multi-step operations.
- Maintained clean Git workflow with structured commits and version-controlled development throughout the internship project lifecycle.

EDUCATION

B.Tech in Computer Science & Engineering (Data Science)

St. Thomas College of Engineering & Technology | KTU, Kerala

Expected Graduation: 2028 | CGPA: 7.23/10

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Hadoop & Big Data, Data Analytics, Cryptography & Network Security, Computer Organization & Architecture